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MARMARA UNIVERSITY FACULTY OF ENGINEERING

EE 2004.1- Microprocessor Systems – Project

Instructor: Dr. Fazlı Kemal Bayat

Transferring CO_2 Data from STM32 to Raspberry Pi Pico W via UART and Displaying It on a Web Interface

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1. Objective

The main objective of this project is to develop a real-time air quality monitoring system that measures CO_2 gas concentration using a sensor connected to an STM32 microcontroller. The collected data is transmitted via UART to a Raspberry Pi Pico W, which then hosts a web interface to display the readings. This setup aims to offer a portable, low-cost, and wireless solution for continuously tracking indoor air quality.

2. System Architecture and Components

The system is composed of several key components that work together to achieve the intended functionality. The MG811 CO_2 gas sensor is used for measuring carbon dioxide concentration in the environment. The STM32 microcontroller reads the sensor data and transmits it using the UART protocol at a baud rate of 9600. The Raspberry Pi Pico W receives the data and acts as a web server to display it to users through a browser-based interface. The software development was carried out using STM32CubeIDE for the STM32 board and Thonny IDE with MicroPython for the Pico W. The user interface was designed with basic HTML and JavaScript, which are served directly from the Pico W.

3. System Workflow

The system operates in a sequential loop to maintain continuous and synchronized data flow. First, the STM32 microcontroller reads CO_2 concentration values from the MG811 sensor at one-second intervals. These values are then transmitted to the Raspberry Pi Pico W over a UART connection. The Pico W receives and stores the most recent value, which is then made available through a simple web server running on the device. A browser-based HTML interface, supported by JavaScript, regularly fetches the latest value from the server and updates the display in real time, providing users with a live view of CO_2 levels.

4. STM32 Code – CO_2 Data Acquisition and Transmission

```
#include "src/CO2Sensor.h"

CO2Sensor MG811(A5, 0.99, 100);
HardwareSerial MySerial(PA9, PA10);

void setup() {
  Serial.begin(9600);
  MySerial.begin(9600);
  MG811.calibrate();
}
```

```
void loop() {
  int co2_val = MG811.read();
  MySerial.println(co2_val);
  Serial.println(co2_val);
  delay(1000);
}
```

5. Raspberry Pi Pico W Code – Web Server and Data Display

```
import network, socket, time, machine, ujson

uart = machine.UART(0, baudrate=9600, tx=machine.Pin(0), rx=machine.Pin(1))
SSID = "AndroidAPC6D8"
PASSWORD = "silver2002"

def connect_wifi():
    wlan = network.WLAN(network.STA_IF)
    wlan.active(True)
    wlan.connect(SSID, PASSWORD)
    while not wlan.isconnected(): time.sleep(1)
def get_html():
    return """<html><head><meta charset='UTF-8'><title>Live CO2</title></head><body><h1
id='co2'>Loading...</h1></body></html>"""
def start_server():
    addr = socket.getaddrinfo('0.0.0.0', 80)[0][-1]
    s = socket.socket()
    s.bind(addr)
    s.listen(1)
    last_value = "N/A"
    while True:
        cl, addr = s.accept()
        try:
            cl_file = cl.makefile("rwb", 0)
            request_line = cl_file.readline()
            request = request_line.decode().split(' ')[1]
            while cl_file.readline() != b'\r\n': pass
            if uart.any(): last_value = uart.readline().decode().strip()
            if request == "/co2":
                cl.send(b"HTTP/1.0 200 OK\r\nContent-Type: application/json\r\n\r\n")
                cl.send(ujson.dumps({"co2": last_value}))
            else:
```

```
            cl.send(b"HTTP/1.0 200 OK\r\nContent-Type: text/html\r\n\r\n")
            cl.send(get_html())
        except: pass
        finally: cl.close()

connect_wifi()
start_server()
```

6. Test Results and Observations

During testing, the UART communication between the STM32 and Raspberry Pi Pico W remained consistently stable at a baud rate of 9600. The system successfully transferred sensor data in real time without experiencing synchronization issues or data loss. The web interface, served directly from the Pico W, displayed the live CO_2 values accurately and updated them once per second as intended. Furthermore, the system was tested across different devices and platforms; both desktop and mobile web browsers were able to access the interface and display the data responsively. The seamless interaction across platforms demonstrated the practicality and portability of the system in real-world applications.

7. Conclusion and Suggestion

This project effectively demonstrated a real-time CO_2 monitoring system using UART communication and a lightweight Wi-Fi-based web interface. The integration of STM32 and Raspberry Pi Pico W allowed for a flexible and low-cost embedded solution suitable for indoor air quality tracking. The system met its design objectives by enabling real-time wireless monitoring and providing reliable performance. For future development, several enhancements can be considered. These include adding historical data logging capabilities to visualize trends over time, integrating MQTT protocol support for IoT cloud communication, and assigning a static IP address to the Pico W for easier and consistent access within a network environment.